

Topic: Maximum Mode Timing Diagrams

Q. Discuss the Memory Read and Write timing diagrams for the 8086 microprocessor operating in Maximum Mode. How does it differ from Minimum Mode?

★ Definition to Remember

In Maximum Mode, the 8086 delegates control signal generation to the **8288 Bus Controller**. Instead of generating RD' or WR' , the CPU outputs early status codes ($\text{S2}'$, $\text{S1}'$, $\text{S0}'$) just before T1. The 8288 decodes these to issue the correct timing signals.

1. The Role of Status Signals

- **Pre-T1 Phase:** Before the T1 clock cycle even begins, the 8086 updates $\text{S2}'$, $\text{S1}'$, $\text{S0}'$ to indicate the upcoming cycle (e.g., 101 for Memory Read, 110 for Memory Write).
- The 8288 uses this early warning to perfectly time the output of ALE , DEN , and commands like MRDC' (Memory Read).

2. Memory Read Timing (Maximum Mode)

- **Pre-T1:** $\text{S2}'$, $\text{S1}'$, $\text{S0}' = 1, 0, 1$ (Memory Read).
- **T1:**
 - CPU places address on AD bus.
 - **8288** pulses ALE high to latch the address.
 - **8288** sets $\text{DT/R}'$ LOW (Receive).
- **T2:**
 - CPU removes address.
 - **8288** asserts MRDC' (Memory Read Command) LOW.
 - **8288** asserts DEN HIGH (active high) to enable transceivers.
- **T3:**
 - Memory places data on the bus. CPU reads it.
 - Status pins ($\text{S2}'$, $\text{S1}'$, $\text{S0}'$) return to $1, 1, 1$ (Idle), signaling the operation is ending.
- **T4:**
 - 8288 deactivates MRDC' and DEN .

3. Memory Write Timing (Maximum Mode)

- **Pre-T1:** $\text{S2}'$, $\text{S1}'$, $\text{S0}' = 1, 1, 0$ (Memory Write).
- **T1:**
 - CPU places address on AD bus.
 - **8288** pulses ALE high.
 - **8288** sets $\text{DT/R}'$ HIGH (Transmit).
- **T2:**

- CPU places data on AD bus.
- **8288** asserts `AMWC'` (Advanced Memory Write) or `MWTC'` (Memory Write) LOW.
- **8288** asserts `DEN` .
- **T3:**
- Data remains stable on bus. Status pins return to `1, 1, 1` .
- **T4:**
- 8288 deactivates `MWTC'` and `DEN` .

4. Key Differences from Minimum Mode

- **Signal Source:** 8288 generates ALE, DEN, DT/R', not the CPU.
- **Command Names:** Uses `MRDC'` / `MWTC'` instead of generic `RD'` / `WR'` .
- **DEN Polarity:** DEN from 8288 is **active-high** (it was active-low `DEN'` in Min Mode).
- **Early Status:** Status pins (`S2-S0`) change before T1 starts.

★ Must-Write Points (for 10 marks)

1. In Max Mode, the **8288 Bus Controller** handles the timing signals.
2. CPU communicates via **S2', S1', S0' status lines**, which change *before* T1 begins.
3. **T1:** 8288 generates **ALE** to latch the address put out by the CPU.
4. **T2:** 8288 asserts specific commands like **MRDC'** (Memory Read) or **MWTC'** (Memory Write) instead of generic RD/WR.
5. **DEN** is generated by the 8288 and is **active-high** in Max Mode.
6. Advanced Write Commands (`AMWC'`) can be used to give slow memory more time to write.
7. In T3, the status lines revert to `1, 1, 1` (Idle) to signal the end of the operation.

⚡ Quick Recall

Max Mode Timing → 8288 in charge → S2,S1,S0 set before T1 → 8288 generates ALE (T1), MRDC/MWTC (T2), DEN (T2) → Status goes idle (1,1,1) in T3 → DEN is active-high